

OpenBCI + 3D-Printed Headphones = Open ExG Headphones – An Open-Source Research Platform for Biopotential Earable Applications

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ABSTRACT

Earables research is a promising field in Human-Computer Interaction (HCI) due to the variety of observable phenomena that can be captured by sensors placed in the ear region. However, this field is hindered by the use of custom-built hardware and software, which can create barriers for less experienced researchers and lead to high assessment heterogeneity. To address this challenge, we have developed a platform for electrophysiological ear-centered research. This platform combines the established amplifier hardware and recording software stack of the OpenBCI biosignal ExG system with a low-cost 3D-printed over-ear headphone design, based on electrode positioning best practices from neuroscientific research. To demonstrate the flexibility and potential of this platform, we have conducted two studies replicating classical EEG, EOG, ECG, and EMG patterns, and showing that the system is reliable and easy to use in a promising application scenario: detecting mental workload levels across multiple tasks.

CCS CONCEPTS

• **Hardware** → Emerging technologies; Biology-related information processing; Neural systems.

KEYWORDS

Earables, Headphones, OpenBCI, EEG, ECG, EOG, EMG, Mental Workload

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1 INTRODUCTION

Ear-centered sensing represents a dynamic and exciting area of research as it opens up a wide range of possibilities for human-computer interaction. Applications such as detecting sleep quality [30], monitoring eating behavior [3], user authentication [8] and new interaction modalities [34] are just a few examples of its potential. The ear region is particularly appealing as it lends itself well to wearable designs and is widely accepted by users. For these reasons, interest in earables has skyrocketed in the recent years [9, 33]. At the same time, this research direction faces high entry barriers that limit the growth of the field. A lot of the current research relies on custom hardware artefacts that require a substantial amount of expertise with electronics design, material science and software engineering [32]. In addition, many of the previous designs focus on few measurement options – which narrows the re-usability of a particular device for other research questions. Furthermore, as many individual solutions do not share common nomenclature, the findings' comparability is limited – what researchers from the earables domain call a problem in assessment heterogeneity [33].

The main goal of our new ExG system design is to lower these entry barriers and provide a multi-purpose platform for earables researchers. ExG, thereby, stands for an amalgamation of multiple electrophysiological measurements like EEG (brain), EMG (muscle), ECG (heart) and EOG (eyes). We leveraged the electronics hardware and software stack of an established open-source project (OpenBCI ExG amplifiers) and adapted an open-source over-ear headphones design to feature up to 21 electrode positions for an integrated, flexible, and standardized ear-centered ExG platform. The system can be replicated by any scholar with a regular desktop 3D-printer, or by ordering the parts from a 3D-printing service at a low price (component total cost is around 100-150 USD; excl. the OpenBCI amplifiers). For the community, this represents an effort to improve the quantity and quality of ear-centered research with a shared methodology that parallels other efforts like open-source in-ear sensing systems (e.g. OpenEarable [35]). It's worth noting that OpenBCI currently lists over 200 publications on their website, which represent just a fraction of the already distributed OpenBCI ExG amplifiers. Thus, our system offers an opportunity for these amplifiers' owners to quickly join the expanding earables research field.

In the following sections we further describe the present state of this field, the Open ExG Headphones design and the results of

two user studies. Altogether, this article, contributes to the earables literature by (1) providing the design and instructions for an over-ear ExG earables research platform, (2) with evidence for the technical capabilities that show the systems recording capabilities for EEG, ECG, EMG, and EOG signals at the same time, and (3) a demonstration of the application of the system for a promising use case: monitoring of mental workload across three tasks. As with all research, this system also still faces some limitations that are discussed in the outlook.

2 RELATED WORK

2.1 Earable Research

The integration of sensing technology into ear- and headphones has garnered significant attention in recent years [9, 33]. This is due to the widespread use of these devices for private audio spaces and the unique opportunities presented by the ear as a sensing location. Various anatomical structures, including facial muscles, brain activity, and the cardiovascular system, can be accessed through the ear. A range of sensing principles has already been applied, including inertial sensing [1], microphones [8] and photoplethysmography [37]. Such sensing data enables a wide range of applications, including health-related and fitness [28], activity tracking [36], human-computer interaction [34] and even user authentication [8]. Besides these sensing principles and applications, the proximity of the ears to various sources of electric potentials makes it a particularly intriguing space for further exploration using electrophysiological methods in different locations in and around the ears. Studies have shown the use of EEG to sense brain activity via custom-fit [14] and one-size-fits-all [27] type of earables with electrodes placed both inside [25] and around the ears [4]. It has been demonstrated that such sensing can support the detection of a wide range of evoked potentials, from audio to visual stimuli [4, 9]. Additionally, EMG has been utilized to detect eating activity [3] and EOG to determine the gaze of the wearer [23]. There is also some preliminary evidence that ECG signals can be sensed by electrodes around the ears [11, 20]. ExG based sensing also affords a series of interesting applications, such as detecting the hearing direction of the user [26], brain-computer interfaces [38], epilepsy detection [10], and sleep tracking [30]. In these studies, many researchers relied heavily on custom-built solutions, lacking a standardized approach for constructing sensing interfaces for biophysiological signals. Earables scholars have, therefore, noted a critical limitation in the field in terms of assessment heterogeneity [33], which we aim to reduce.

2.2 Earable Development Platforms

Recent developments in earable research have seen an increase in the availability of development platforms. Commercial platforms, such as Apple's AirPods, provide access to the IMU for motion data sensing. Additionally, research-specific standard platforms, such as eSense [16], have emerged and generated a substantial body of literature. Furthermore, DIY open-source development earbud platforms, such as ClearBuds [5] and OpenEarable [35], have also appeared in the field. These devices primarily focus on sensing in-ear signals, such as motion data, ear canal pressure, and audio inside and outside the ear canal. On the over-ear side, a commercial platform for EEG sensing are the SMARTFONES by mbraintrain,

which directly integrate electrodes into a headphone form factor [15] and cEEGrid electrodes, which are flexible PCB electrodes that wrap around the ears by sticking them onto the skin [4, 7]. While these two approaches provide solutions for around-the-ear and top-of-the-head sensing, they require very costly amplifiers and offer only a subset of useful electrode positions. Thus, a central gap for a versatile headphones system that can be used to study a wide variety of phenomena in the face and ear region remains. Importantly, while over-ear designs are more visually salient, they offer essential advantages for ExG sensing, like larger inter-electrode distances (for capturing small-amplitude phenomena) and higher sensor proximity to various electrical sources (facilitating the isolation of these sources). For these reasons, we believe that an over-ear ExG system possesses essential characteristics for a valuable research platform.

3 SYSTEM DESCRIPTION

To develop the Open ExG headphone system, two available open-source projects were adapted and joined. Importantly, this means that the core of the measurement system is built on established technology, the amplifiers, electrodes, and software stack of the OpenBCI biosignal acquisition platform. OpenBCI offers three amplifier alternatives, the Ganglion, Cyton, and Cyton+Daisy boards, which allow for 4, 8, or 16 ExG channel recordings, with sampling rates of 200, 250, and 125Hz, respectively. In this article, we primarily use the Cyton and Cyton+Daisy combination (as we used many recording channels simultaneously) that feature 24-bit resolution, low input-referred noise and low power consumption [32]. With a 1000mAh LiPo battery and a 16GB SD card continuous recordings of over 10 hours can be achieved [17]. Besides the amplifiers, OpenBCI provides metal Ag/AgCl electrodes (that we re-use), snap button cables for ExG recordings (that can be used to collect reference recordings with pre-gelled electrodes), and a software stack (GUI and APIs) for easy recordings and application development. Normally, this amplification system is used with a 3D-printed head harness, the Ultracortex MK IV (see Figure 1A), that positions electrodes across the head in a classical EEG layout – the 10-20 system [29].

Our ExG headphone system, in contrast, allows ExG recordings on five typical electrode positions on the top of the head (T3, C3, Cz, C4, T4) and 16 around the ear (L1-L8 + R1-R8), featuring about the same electrode positions that are recorded with the around-the-ear cEEGrid electrodes [7]. We consider these cEEGrids as an established reference system for around-the-ear EEG electrode positions as it has repeatedly been used in neuroscientific EEG research [4, 7]. Thereby, our system's design is also rooted in established electrode position nomenclatures (see Figure 2).

Lastly, by building on the design from the "Print.Plus" open-source headphones project, we also leveraged established aesthetic and comfort factors for the headphones design. Thereby, the components can be produced by fused-deposition modeling (FDM) desktop 3D-printers. In addition, the headphones are fitted with audio drivers to feature a fully functional headphones system with speakers and a microphone. All the components can be freely purchased and assembled at low cost (~60 USD for the components + 40 USD for filament or + 100 USD using a printing service). The materials

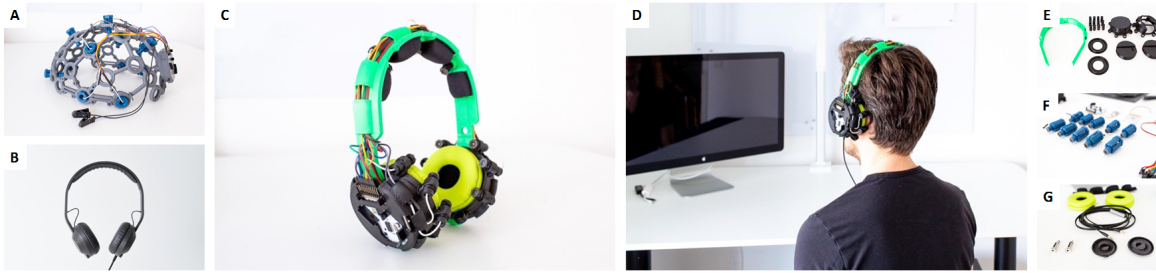


Figure 1: Genesis and components of the Open ExG Headphones. Adaptation of the source material (A & B) led to the headphone design (C & D). E-G show the individual components: 3D-printed parts (E), OpenBCI Electronics (F), and audio components (G).

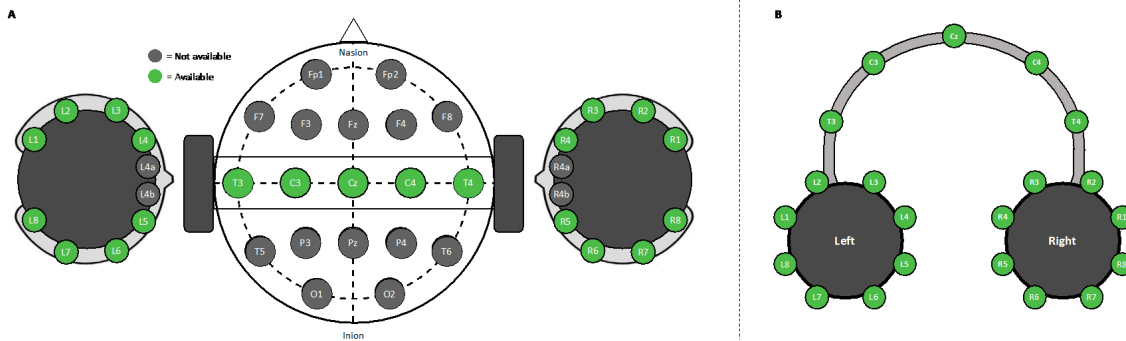


Figure 2: A) Standardized electrode positions in the EEG headphones system. The layout combines the 10-20 electrode positioning system for EEG measurements [33] and the positioning reference of around-the-ear cEEGrid electrode arrays [9]. B) A simplified electrode position layout that is used for the studies in this article.

for building our system (list of components and assembly instructions) are provided here: <https://github.com/MKnierim/openbci-headphones>.

In summary, our system features three key attributes for advancing earables research: (1) configurability (selectable electrode positions and open-source modifiability of all hardware and software components), (2) standardized sensor positions (allowing for comparable study results), and (3) availability through easy self-assembly at low cost (facilitating research participation even with smaller budgets).

4 STUDY 1: DEMONSTRATION OF EXG SIGNAL RECORDINGS

4.1 Study Protocol & Materials

To demonstrate the versatile capabilities and the configurability advantage of the ExG headphones, a first series of measurement demonstrations was recorded with electrode positions for the capture of classical brain (C3, C4, L4, R4), heart (L5, L6, R5, R6), face muscle (L1, L8, R1, R8), and eye activation (L1, L8, R1, R8) patterns. Concurrently gold-standard reference recordings were recorded with pre-gelled adhesive electrodes on bipolar channels for EOG (four electrodes close to the eyes on the left and right temple, and above and below the left eye, with a ground electrode on the top of the forehead), EMG (two electrodes in a vertical arrangement on the masseter muscle on the cheek, also using the forehead ground),

and ECG (two electrodes on the left and right chest, with a ground electrode on the left hip). The entire configuration and placements are visualized in Figure 3A.

To record the activation patterns, one participant (male, 34 years) was instructed to (1) sit comfortably with eyes opened and closed for 90s (to produce the Alpha band power increase in the EEG when the eyes are closed - [2]), (2) to blink, follow a moving dot with his eyes from left to right for ten times, and (3) chew gum for 15s (see Figure 3B).

4.2 Data Processing & Results

To assess the *EEG recording* capabilities by producing the Berger effect, the recorded signals were mean-centered and band-pass filtered (2-30Hz, FIR) and the power spectral distribution (PSD) was extracted in two ways similar to the related work [20]. First, using the Welch-windowed short-term Fourier Transform (2s windows with 50% overlap), the PSD was extracted for the entire eyes-open and eyes-closed segments and averaged across the electrodes. Figure 4A shows the characteristic elevation in the Alpha frequency band (8-12 Hz region) during the eyes closed condition (in a dB-normalized PSD in the top row and a PSD normalized by total power division in the bottom row). Second, to study the temporal characteristics of this effect, a time-frequency PSD extraction using multitapers was performed. Figure 4B also shows the elevation

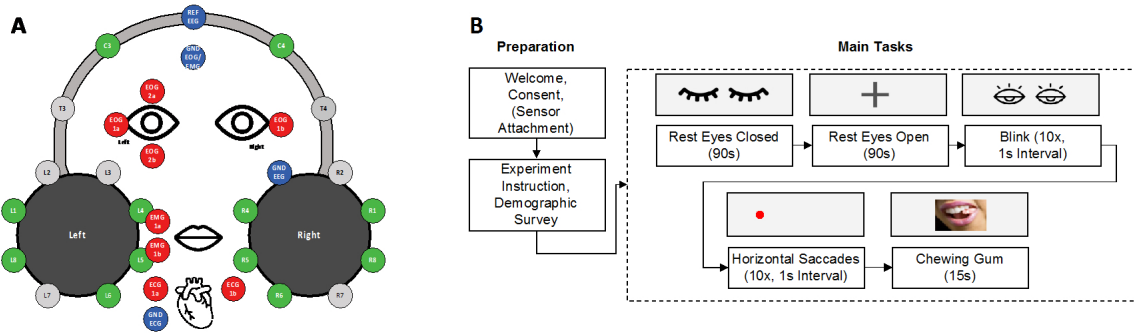


Figure 3: A) Electrode configuration and B) protocol for the ExG signal demonstration study.

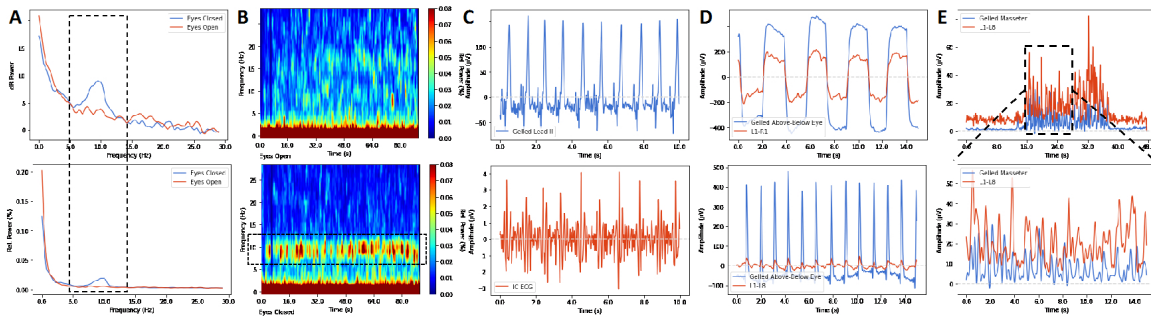


Figure 4: A) Average power distributions during rest with eyes open and closed. Both dB normalized (top) and total-power normalized (bottom) PSDs are shown. B) Time-Frequency average power distributions during the eyes open and closed rest conditions. The PSDs are total-power normalized. C) Comparison of standard Lead II ECG (top) and IC-extracted ECG trace (bottom). D) Comparison of standard and headset EOG signals showing horizontal eye movements (top) and blinks (bottom). E) Comparison of standard and headset masseter EMG signals for the entire chewing segment (top) and a 15s extract (bottom).

of the Alpha frequency band power continuously throughout the eyes-closed setting.

To demonstrate the **ECG recording** capabilities, we initially pursued a simple approach using the homologous electrode pair behind the ear (R5-L5 and R6-L6) to compute ECG channels, similar to previous work [20]. The signals were band-pass filtered (5-15Hz, FIR – as is recommended in ECG signal analysis - [22]). This approach did not produce a clearly visible ECG trace. We believe this is likely caused by too high electrode-skin impedances. However, we identified a way to extract a clear ECG signal by computing an independent component analysis (ICA - method: fastica) on the four electrodes behind the ear (L5, L6, R5, R6). Figure 4C shows the IC-extracted ECG in comparison to the standard Lead II Einthoven ECG with self-adhesive gelled electrodes.

To investigate the detection of **EOG signals**, the recordings during the blink and horizontal eye movement stage were mean-centered and band-pass filtered (0.25-6.5 Hz FIR) and horizontal (L1-R1) and vertical (L1-L8) EOG channels computed by subtracting the latter channel from the former. Figure 4D highlights the similarity of the traces in comparison to standard hEOG and vEOG channels with self-adhesive pre-gelled electrodes. It should be noted that the signal amplitudes are much smaller on the headphone channels, which is caused by the larger distance from the signal source.

Finally, to assess the detection of facial **EMG signals**, the chewing data segments were processed following related EMG work [31]: after channel computation (the same vertical L1-L8 channel as for the EOG analyses), the signals were band-pass filtered (7-62.5 Hz, FIR), rectified (taking the absolute values of the signals), and finally low-pass filtered (3Hz, FIR) to emphasize the slope of the signal. Figure 4E shows the strong masseter muscle activations that are actually more pronounced in the headset channel than the reference channel. This is likely caused by the additional electrode movement that is not similarly present in the gelled adhesive electrodes.

Altogether, these results demonstrate the potential of our ExG headphones for collecting EEG, ECG, EOG, and EMG data. To further assess the system's application potential, we decided to also conduct a larger study that focuses on a particular advantage of the over-ear headphones design, that is the monitoring of mental workload which should be especially supported by collection of EEG data on the top of the head (see e.g. [15] and [6] which show that mental workload changes are particularly visible on the center of the head in EEG recordings).

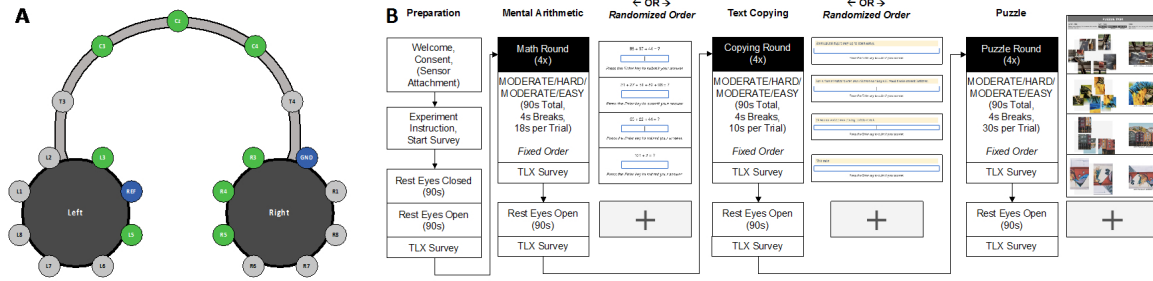


Figure 5: A) Electrode configuration and B) protocol for the workload monitoring study.

5 STUDY 2: MENTAL WORKLOAD DETECTION ACROSS TASKS

5.1 Study Protocol & Materials

For the workload monitoring evaluation, headset electrode positions were chosen that were expected to most strongly reflect workload changes: positions on top of the head (C3, Cz, C4) and towards the occipital cortex behind the ears (L3, L5, R3, R4, R5) – see Figure 5A. Only the Cyton Amplifier was used for faster setup and a higher sampling frequency.

We implemented an experiment that manipulates task difficulty in mental arithmetic, puzzle, and text copying tasks, with three difficulty levels (easy, medium, and hard) – see Figure 5B. These three tasks were selected to provide variation in mental, visual, and motoric demands, which are known as confounds of mental workload correlates [6]. In the mental arithmetic task, participants had up to 18s to add a manipulated number of summands (2, 3, and 5 summands in the easy, medium, and hard task conditions). In the puzzle task, participants had up to 30s to assemble a picture puzzle with a manipulated number of pieces (2, 6, and 12 pieces in the easy, medium, and hard task conditions). In the text copying task, participants had up to 10s to transcribe the text from a picture with a manipulated number of characters (9-10, 31-35, and 61-65 characters in the easy, medium, and hard task conditions). All task rounds lasted 90s, with 4s breaks between two consecutive stimuli. After an introductory task round in medium difficulty, participants completed a hard, then a medium, and lastly, an easy difficulty task round. The order of the three tasks was randomized. Each task round was followed by a short questionnaire, including the NASA TLX [12], to assess the perceived mental workload.

Participants were recruited from a student participant pool at the first author’s university. Overall, 18 participants (6 female, age range = 21-34, median age = 27, 9 wearing glasses during the experiment) completed this experiment. One participant was excluded as a power outage in the experiment PC terminated the experiment early. We tracked the setup time from putting the headphones on to the start of the experiment to quantify some of the system’s usability aspects. The median setup duration (including signal quality inspection) was 4.14 minutes (min = 1.11 minutes, max = 13.91 minutes). Wilcoxon rank test results indicate a significant difference in setup time for gender ($W = 60$, $p = .0245$ – possibly caused by longer hair) and no significant difference in setup duration for participants wearing glasses ($W = 49$, $p = .4894$).

5.2 Data Processing & Results

For the mental workload classification, the TLX prediction labels were created by summing the responses on the six TLX items and subsequent z-score normalization. Afterwards, these values were binned into three classes (low, medium and high load). On top of that, a fourth class was added (eyes open rest) to assess if rest (no load) can be differentiated from low load situations. We used the StratifiedKFold cross-validation strategy to split the data into five folds.

For the classifier, we pursued a strategy with minimal EEG data processing to evaluate the potential of the headphones’ data “out-of-the-box”. The EEG signals were only re-referenced to a common average montage and band-pass filtered (1-15Hz, FIR). Next, we used a specific convolutional neural network (CNN) that has been developed for fast use with EEG signals: EEGNet [19]. EEGNet provides a lightweight and efficient CNN architecture consisting of a 1D-temporal convolutional layer that extracts feature maps at different band-pass frequencies for each channel, followed by a depthwise spatial convolutional layer that enables the network to learn spatial filters for each extracted temporal filter. To enable classification, the continuous EEG signals were divided into 3.13s long segments with 90% overlap (resulting in 3600 windows per participant). This timeframe resulted from the tuned window width of 800 datapoints which were recorded with a sampling rate of 256Hz. As the mental workload reports of the participants are subjective, it is not guaranteed that the classes in the data are balanced. To counteract this, windows belonging to underrepresented classes were oversampled. Hyperparameters for the EEGNet and the data pre-processing were identified by optimizing testing accuracy using the Optuna Search Algorithm implemented in the Ray Tune framework [21]. The search was conducted for a subset of three participants. The specific hyperparameters used for training the model were: 400 epochs, a learning rate of 0.00505, a dropout rate of 0.5, a number of spatial filters D of 2, a filter size of 12 and a kernel length of 140 and a batch size corresponding to batch gradient descent. The EEGNet was trained using the Tensorflow framework with ADAM as an optimizer and Sparse Categorical Crossentropy as a loss function.

Our subject-dependent models show an average test accuracy of 70%. However, there is a significant variation in performance among subjects, as shown in Figure 6A. The lowest-performing model has an accuracy of 55%, while the highest-performing model has an accuracy of 85%. For comparison, a dummy classifier achieves 25%

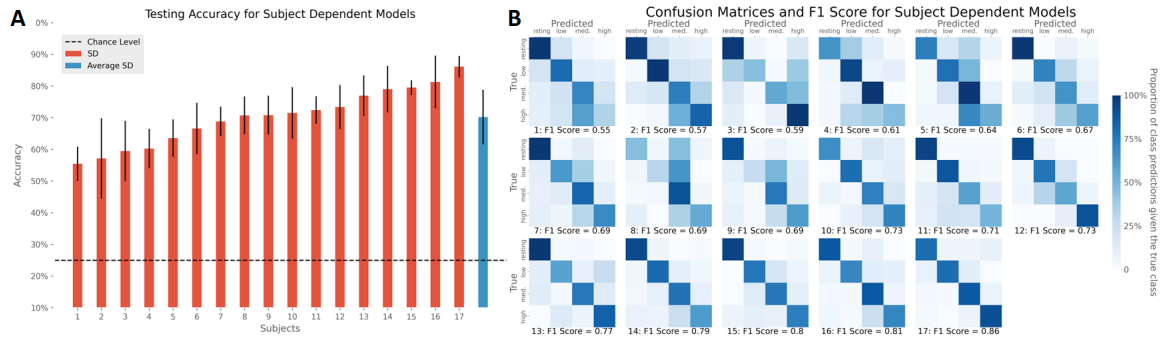


Figure 6: A) Average testing accuracies for subject dependent models. The error bars represent the standard deviation of the folds. B) Confusion matrices and F1 score for subject dependent models.

accuracy by random chance on four equally distributed workload classes. Figure 6A shows the accompanying confusion matrices and weighted F1 scores for each participant in the same order. The weighted F1 scores are similar to the reported accuracies, and only next-door classes show higher confusion on occasion. The better-performing subject models have consistent performance across all classes rather than significantly higher per-class accuracies.

6 DISCUSSION AND CONCLUSION

With this article, we aim to demonstrate the recording capabilities of a novel ExG headphones platform that flexibly allows studying brain, heart, muscle, and eye activation patterns with standardized electrode positions and building on an established hardware and software stack. The general propensity for the system to collect EEG, EMG, and EOG patterns was qualitatively demonstrated by replicating classical phenomena with a single subject. Certainly, more extensive evaluations will have to follow to assess the boundaries for such recordings. For example, it will have to be determined which level of sensitivity can be achieved in contrast to gold standard references and how robust the respective recordings are in the presence of environmental and bodily noise sources. This holds especially true for the collection of ECG traces, which were found to be possible only through independent component analysis. This finding was a bit surprising as previous work showed clearer ECG traces from electrodes in similar positions [20]. It is possible that the skin-electrode impedances were too high in our recording, as a later, second recording showed that gelled electrodes in the same position showed more prominent ECG traces by mere left-right channel subtraction (i.e. using one electrode as the reference channel for the other one). Thus, when a real-time signal feed is required, a simple solution for the collection of ECG traces in the headphones system could be adding a drop of electrolyte gel on an electrode pair behind the ear. However, for offline analyses with sufficiently large observation segments (e.g. >60s), ICA decomposition also allows for clean ECG extraction.

Beyond this demonstration of ExG recording properties, our second experiment showed very good results for monitoring mental workload near real-time with subject-dependent classification accuracies (and four predictable classes) reaching up to 85% accuracy. For comparison, other wearables have achieved similar classification

accuracies, but for only two predictable classes (see, e.g. [13] with a forehead-electrode headband and load classification accuracies of 69%, or [20] with an average classification accuracy of up to 79% with in-ear sensors). Stated simply, our classification results mean that no, low, medium and high levels of workload can be predicted well across three different tasks in near real-time. We believe this to be a promising direction for further utilization of the system, even in a field study, especially as prolonged and extreme mental workload has been linked to detrimental health consequences in the work context [24]. If these workload prediction accuracies can be maintained in field settings (e.g. in knowledge work conducted in a quiet room – a home-office scenario), they would lend a useful data basis for health-supporting feedback systems. Together with that, it would be interesting to see how facial muscle activations and eye activity patterns can be integrated into such recordings, on the one hand for artefact removal, but also to study specific behavioral patterns (emotion expression, bruxism, eating, fatigue via blinking – see e.g. [18] for related evidence). We believe our system to be closely ready for such application scenarios, especially as we observed fast setup times and good usability across gender (hairstyles) and glasses. However, personal experiences in the experiment setup and feedback from study participants also lead us to believe that further improvements can be made to the headset, for example, in terms of comfort and electrode placement reliability in the ear region. A way to achieve both would be to convert the current on-ear cushion placement to an over-ear design, where the electrodes are integrated with the supportive cushion. That way, both increased comfort and fastened electrode positions could be achieved. Importantly, as our described system is entirely available under open-source licenses, others can either use or adapt the system and contribute to developing the platform for the earables community. Together with the various application possibilities that it already offers and the use of widely distributed biosignal amplifiers, we believe it can valuably contribute to further advancing and consolidating the great potential in the earables space.

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